THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Earth's axis is tilted by about: 1

(a) 0° **(b)** 90° **(c)** 45° **(d)** 23.5°

2. When it is summer in the northern hemisphere, it is ___ in the southern hemisphere. 1

(a) the same **(b)** summer **(c)** spring only **(d)** winter

3. After full moon, the Moon's bright part: 1

(a) decreases (waning) **(b)** increases
(c) doubles **(d)** stays same

4. A safe way to see a solar eclipse is through: 1

(a) naked eye **(b)** binoculars without filter
(c) a pinhole projection or special filters **(d)** sunglasses

5. The imaginary line dividing the Earth into northern and southern hemispheres is the: 1

(a) axis **(b)** equator **(c)** prime meridian **(d)** tropic of Cancer

6. In summer, the Sun's rays are more: 1

(a) reflected **(b)** direct (vertical)
(c) absent **(d)** slanting

7. The Moon shines because it: 1

(a) is hot **(b)** reflects sunlight
(c) makes its own light **(d)** burns

8. The Moon's dark central shadow in a solar eclipse is the: 1

(a) corona **(b)** halo **(c)** umbra **(d)** penumbra

9. The year in which February has 29 days is called a: 1

(a) normal year **(b)** lunar year **(c)** leap year **(d)** solar year

10. In winter, the Sun's rays are more: 1

(a) vertical **(b)** bright **(c)** absent **(d)** slanting

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why should one not look at the Sun directly during eclipse? 2

12. What is a leap year? 2

13. Why do we see only one side of the Moon? 2

14. What is the Moon's time of revolution? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the Moon's phases occur. 3

16. Why does the Sun appear to rise in the east and set in the west? 3

17. What happens at the equator regarding day length? 3

OR

How does a solar eclipse occur? 3

18. Describe how to observe a solar eclipse safely using a pinhole. 3

OR

Explain why the Pole star appears fixed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the equinoxes and solstices with dates and day-night lengths. 5

OR

Describe an activity to show the cause of day and night using a globe and a torch. 5

20. Explain why the seasons are not caused by the Earth's distance from the Sun. 5

OR

Describe an activity to model the phases of the Moon with a ball and a lamp. 5

END OF QUESTION PAPER — Total: 40 Marks

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